MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 1, 2025/2026

PROPOSAL FOR WASHING MACHINE PROTOTYPE

SECTION 1

GROUP 20

LECTURER: DR. WAHJU SEDIONO

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ABSTRACT

This project presents the design and development of a washing machine prototype that demonstrates basic automation, safety control, and sensor integration using an Arduino-based system. The prototype is intended to simulate the operating logic of a real washing machine. Various sensors and output components are integrated to ensure safe operation and clear user feedback. An IR sensor is used to detect the door status, an ultrasonic sensor is used to estimate the laundry load, and a DHT11 sensor monitors temperature to prevent overheating. A Pixy camera is implemented to detect white cloth, while an HC-05 Bluetooth module enables wireless communication. System messages and operating conditions are displayed using an LCD.

A DC motor is used to represent the washing drum operation. Visual and audio indicators are provided through a green LED, a red LED, and a buzzer. The green LED indicates normal operation, while the red LED and buzzer are activated during unsafe conditions such as an open door, overload, or excessive temperature. The system follows a predefined flowchart that prioritizes safety checks before allowing the motor to operate. This prototype demonstrates how multiple sensors and actuators can be integrated into an automated system and provides a practical learning platform for understanding embedded system control in household appliances.

INTRODUCTION

Washing machines are commonly used household appliances that rely on automated control systems to operate safely and efficiently. These systems use sensors to monitor operating conditions and indicators to provide feedback to users. Understanding how these components work together is important for students studying electronics, instrumentation, and embedded systems.

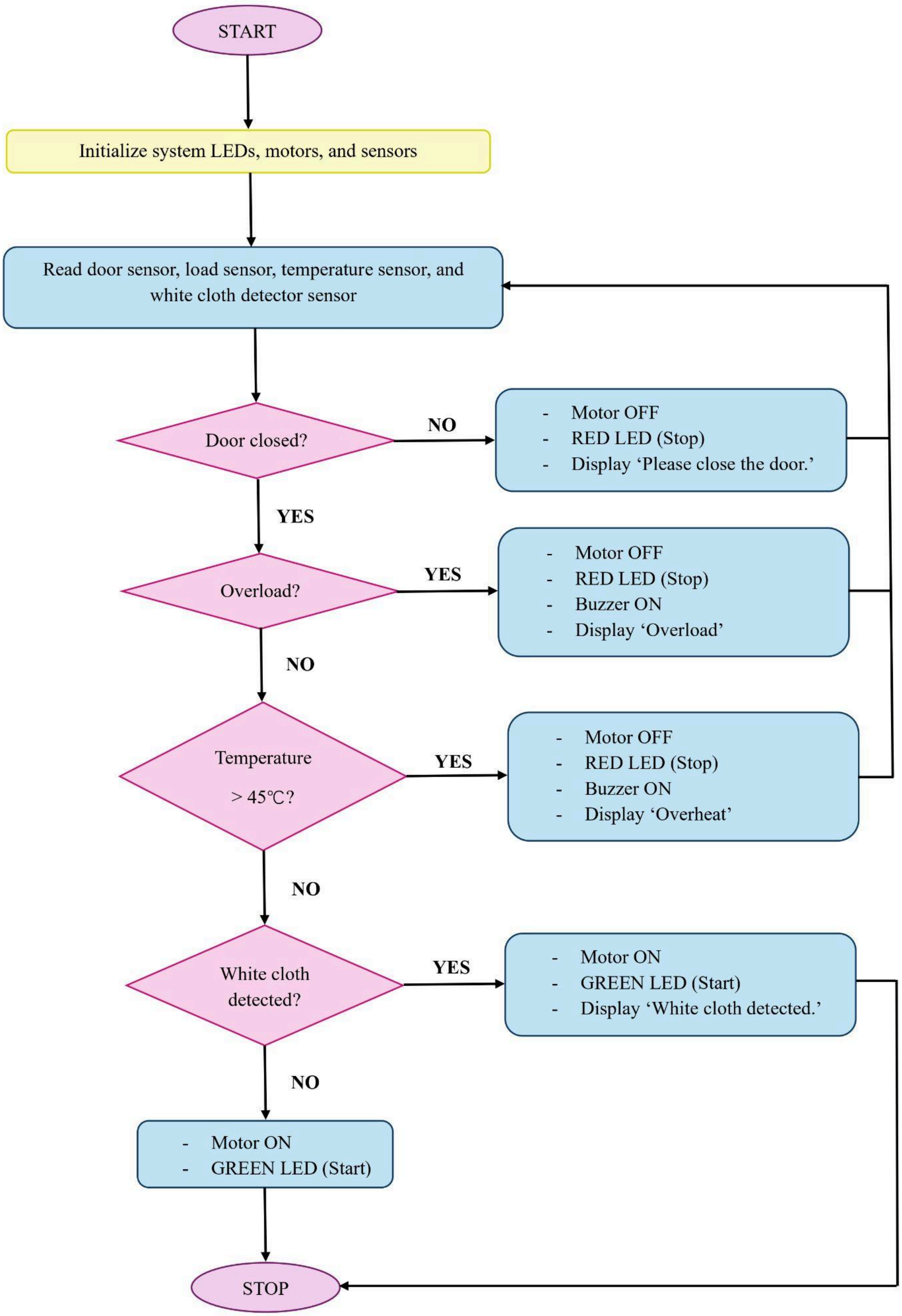
This project involves the development of a washing machine prototype that simulates the basic operation of an automated washing system. The prototype does not involve actual water usage and focuses on control logic and safety-based decision making. An IR sensor is used as a door sensor to ensure that the machine does not operate when the door is open. An ultrasonic sensor is used to estimate the laundry load and detect overload conditions. Temperature monitoring is carried out using a DHT11 sensor to prevent overheating. In addition, a Pixy camera is used to detect white cloth, demonstrating the application of basic machine vision. Wireless communication is supported using an HC-05 Bluetooth module, and an LCD display provides real-time system messages and instructions to the user.

A DC motor is used to represent the washing drum in the prototype. System status is indicated using a green LED for normal operation and a red LED for stop or fault conditions. A buzzer is included to provide an audible alert during unsafe situations such as overload or overheat. The overall operation of the prototype follows a structured flowchart where safety checks are performed before the motor is activated. This project highlights the importance of safety, user feedback, and control logic in automated appliance systems and serves as a practical learning tool for embedded system applications.

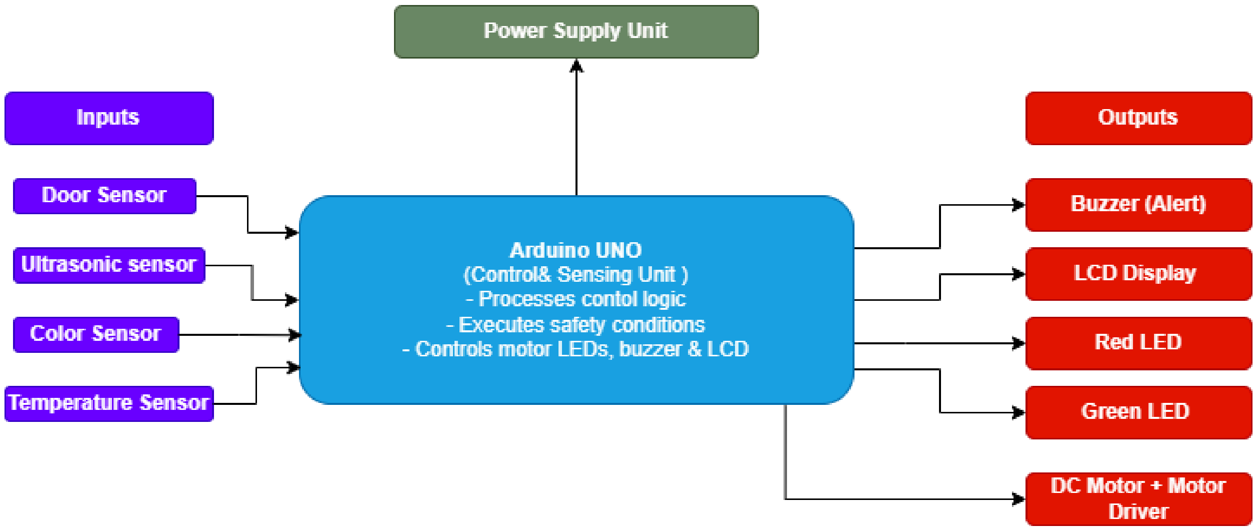
MATERIALS AND EQUIPMENT

1. Arduino Uno
2. IR Sensor
3. Ultrasonic sensor (load sensor)
4. Color Sensor
5. Buzzer
6. DHT11 (temperature sensor)
7. Motor driver
8. Arduino LCD display
9. Green LED
10. Red LED

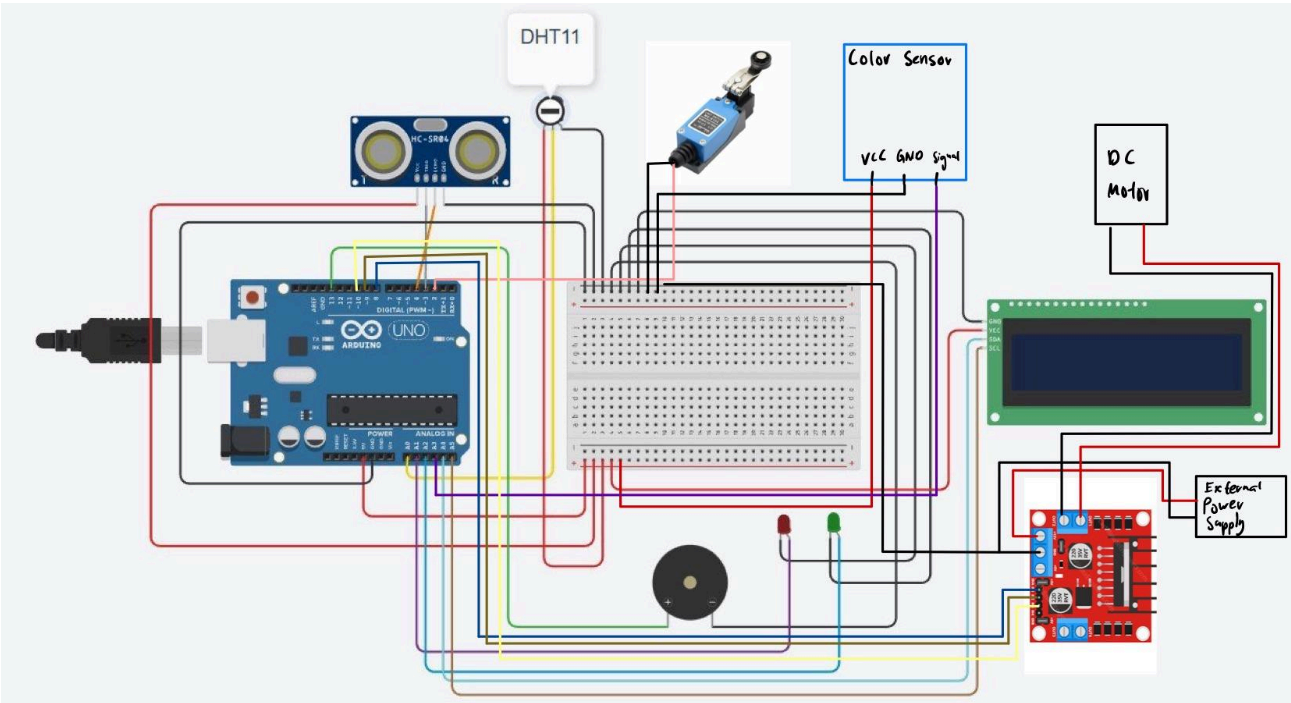
FLOWCHART


START
Initialize system LEDs, motors, and sensors
Read door sensor, load sensor, temperature sensor, and white cloth detector sensor
Door closed?
NO
- Motor OFF
- RED LED (Stop)
- Display ‘Please close the door.’
YES
Overload?
YES
- Motor OFF
- RED LED (Stop)
- Buzzer ON
- Display ‘Overload’
NO
Temperature > 45℃?
YES
- Motor OFF
- RED LED (Stop)
- Buzzer ON
- Display ‘Overheat’
NO
White cloth detected?
YES
- Motor ON
- GREEN LED (Start)
- Display ‘White cloth detected.’
NO
- Motor ON
- GREEN LED (Start)
STOP

SYSTEM ARCHITECTURE



WIRING DIAGRAM



Certificate of Originality and Authenticity

We hereby certify that we are responsible for the work presented in this report. The content is our original work, except where proper references and acknowledgements are made. We confirm that no part of this report has been completed by anyone not listed as a contributor. We also certify that this report is the result of group collaboration and not the effort of a single individual. The level of contribution by each member is stated in this certificate. Furthermore, we have read and understood the entire report, and we agree that no further revisions are required. We collectively approve this final report for submission and confirm that it has been reviewed and verified by all group members.

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